SHOPSPHERE ENGINEERING

API SPECIFICATION

Payment API Specification v2.0

POST /v2/payments: request schema payment-request/2.0, deprecated fields, error codes

Document number	SS-API-PAY-002
Spec version	2.0
Published	2026-08-14
Owner	Payments Team
Applies to	payment-service 2.x
Schema	payment-request/2.0
Clients	order-service (currently the only client)

Fictional document. ShopSphere, its services, people and data are synthetic. This document is part of the TraceIQ training dataset and is not documentation for any real system.

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1. Overview

payment-service exposes one endpoint for creating payments. All clients (currently only order-service) must send JSON bodies that follow the payment-request/2.0 schema described here.

1.1 Document Summary

Item	Value
Spec version	2.0
Published	2026-08-14
Owner	Payments Team
Applies to	payment-service 2.x
Endpoint	POST /v2/payments
Request schema	payment-request/2.0
Success status	HTTP 201 Created
Clients	order-service (currently the only client)

1.2 Conventions

Topic	Convention
Format	Request and response bodies are JSON (Content-Type: application/json).
Money	Amounts are integers in minor units (cents). 4999 = 49.99.
Currency	ISO 4217 three-letter code, for example EUR.
Required	"Required: yes" means the request is rejected if the field is missing.
Errors	Every error response contains error_code, message and trace_id (Section 5).

1.3 Where payment-service Sits

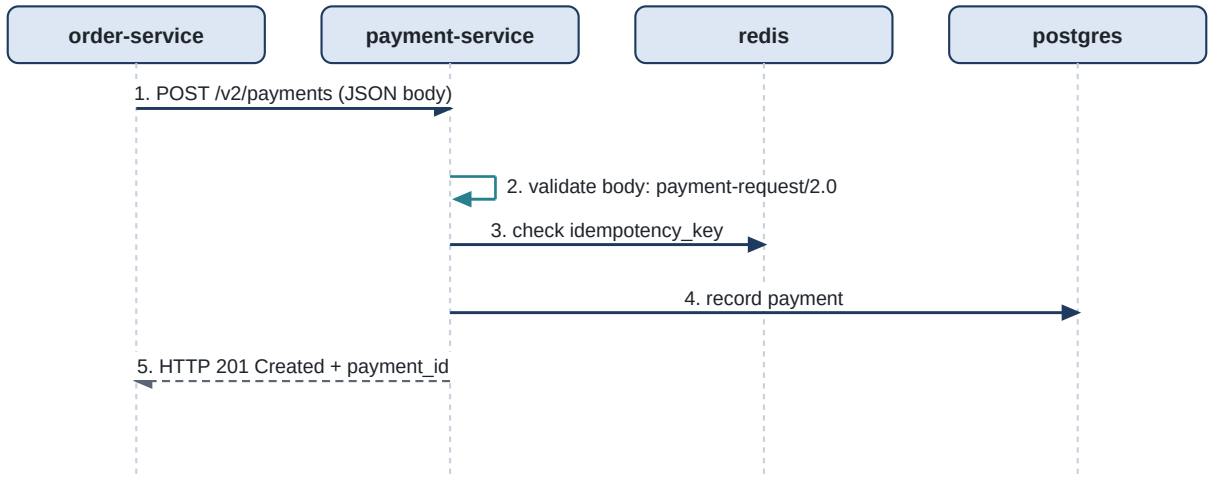

Figure 1-1. A successful call to POST /v2/payments (dashed arrow = response).

2. Endpoint

POST /v2/payments - Content-Type: application/json. A successful payment returns **HTTP 201 Created** with the payment_id.

2.1 Request Line and Headers

Header	Required	Value
`Content-Type`	yes	application/json
`Idempotency-Key`	no	May be sent by the client for logging. The body field idempotency_key (Section 3) is the value that payment-service uses.

Table 2-1. Request headers.

Request line

```
POST /v2/payments HTTP/1.1
Host: payment-service
Content-Type: application/json
```

2.2 Successful Response

When the payment is recorded, payment-service returns HTTP 201 Created. The body contains the payment_id of the new payment and echoes the order_id.

Example: successful response (illustrative values)

```
HTTP/1.1 201 Created
Content-Type: application/json

{
  "payment_id": "pay_000001",
  "order_id": "ORD-10001",
  "status": "captured"
}
```

2.3 Authentication

Calls from order-service to payment-service are service-to-service calls authenticated with a service API key that is kept in the secret store. Authentication is checked before the body is read. Credential problems are therefore separate from body validation and are not part of the PAY error code table in Section 5.

3. Request Fields (schema payment-request/2.0)

Every request body must follow schema payment-request/2.0. The table lists all fields.

Field	Type	Required	Description
order_id	string	yes	ShopSphere order ID, e.g. ORD-88213.
amount_minor	integer	yes	Amount in minor units (cents). 4999 = 49.99.
currency	string	yes	ISO 4217 code, e.g. EUR.
payment_method_token	string	yes	Token for the customer's payment method. Never a raw card number.
idempotency_key	string	yes	Unique per order. Reused on retries.
customer_ref	string	no	Optional customer reference.

Table 3-1. Fields of payment-request/2.0.

Example body: {"order_id": "ORD-10001", "amount_minor": 4999, "currency": "EUR", "payment_method_token": "pm_tok_example", "idempotency_key": "idem-ORD-10001"}

Example: the same body, formatted

```
{
  "order_id": "ORD-10001",
  "amount_minor": 4999,
  "currency": "EUR",
  "payment_method_token": "pm_tok_example",
  "idempotency_key": "idem-ORD-10001"
}
```

3.1 Field Rules

Field	Rule
order_id	Required string. The ShopSphere order the payment belongs to, e.g. ORD-88213.
amount_minor	Required integer in minor units (cents). 4999 means 49.99. Send a whole number, not a decimal and not a string.
currency	Required string. An ISO 4217 code such as EUR.
payment_method_token	Required string. A token that stands for the customer's payment method. It is never a raw card number.
idempotency_key	Required string. Unique per order. When a client retries a payment it must reuse the same key, so that the payment is never recorded twice.
customer_ref	Optional string. A customer reference. It can be left out.

Table 3-2. Rules for each field.

3.2 Required Fields at a Glance

A request passes field validation only when all five required fields are present with the right types:

- order_id (string)
- amount_minor (integer)
- currency (string)
- payment_method_token (string)
- idempotency_key (string)

customer_ref is the only optional field. Deprecated field names are handled as described in Section 4.

3.3 Processing Order

payment-service checks a request in a fixed order. The first check that fails decides the error code.

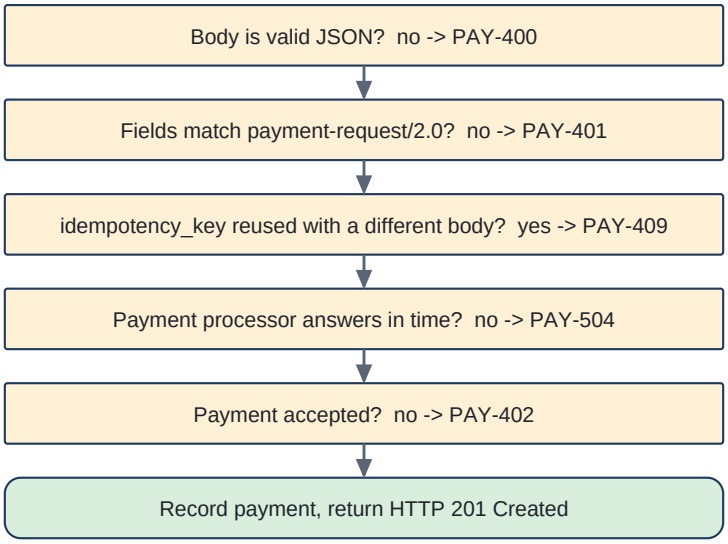

Figure 3-1. Order of checks and the error code returned at each step.

3.4 Examples of Accepted and Rejected Bodies

The cases below show which error code a body produces. Values are illustrative.

Case	What is wrong	Result
All five required fields present with correct types	Nothing	HTTP 201 Created
Body is cut off and is not valid JSON	Not valid JSON	PAY-400, HTTP 400
payment_method_token left out (and no card_token)	Required field missing	PAY-401, HTTP 422
amount_minor sent as the string "49.99"	Wrong type (must be an integer)	PAY-401, HTTP 422
currency left out	Required field missing	PAY-401, HTTP 422
Same idempotency_key as an earlier request, different amount_minor	Idempotency key reused with a different body	PAY-409, HTTP 409
card_token sent instead of payment_method_token	Deprecated alias (Section 4)	Accepted until 2027-03-31: card_token is treated as payment_method_token

Table 3-3. Example bodies and results under this specification.

Example: rejected with PAY-401 because amount_minor is a string

```
{
  "order_id": "ORD-10003",
  "amount_minor": "49.99",
  "currency": "EUR",
  "payment_method_token": "pm_tok_example",
  "idempotency_key": "idem-ORD-10003"
}
```

4. Deprecated Fields

Version 1 of the API used the field name **card_token**. In schema payment-request/2.0, **card_token** is a **deprecated alias** of **payment_method_token**. If a request contains card_token and no payment_method_token, payment-service treats card_token as payment_method_token. The alias will be **accepted until 2027-03-31**. Clients should move to payment_method_token before that date.

Deprecated field	Replaced by	Accepted until
`card_token`	`payment_method_token`	2027-03-31

Table 4-1. Deprecated fields in payment-request/2.0.

4.1 Example: Request Using the Deprecated Alias

The body below uses the version 1 field name. Under this specification it is accepted until 2027-03-31, because card_token is treated as payment_method_token.

Example: body with deprecated card_token (illustrative values)

```
{
  "order_id": "ORD-10002",
  "amount_minor": 2599,
  "currency": "EUR",
  "card_token": "pm_tok_example",
  "idempotency_key": "idem-ORD-10002"
}
```

4.2 Migrating a Client

To move a client to the current field name, rename one field. Nothing else in the body changes.

Before (version 1 name)	After (payment-request/2.0 name)
`"card_token": "pm_tok_example"`	`"payment_method_token": "pm_tok_example"`

Deadline. Clients should move to payment_method_token before 2027-03-31. After that date card_token will no longer be accepted as an alias.

5. Error Codes

Errors return a JSON body with error_code, message and trace_id.

Error code	HTTP status	Meaning
`PAY-400`	400 Bad Request	Body is not valid JSON.
`PAY-401`	422 Unprocessable Entity	Request validation failed (for example a required field is missing or has the wrong type).
`PAY-402`	402 Payment Required	Payment was declined.
`PAY-409`	409 Conflict	Idempotency key reused with a different body.
`PAY-504`	504 Gateway Timeout	Payment processor did not respond in time.

Table 5-1. Error codes of POST /v2/payments.

5.1 Error Response Body

Field	Type	Description
`error_code`	`string`	One of the codes in Table 5-1.
`message`	`string`	Short human-readable description of the error.
`trace_id`	`string`	ID of the request trace. Use it to find the matching log records.

Example: validation error (illustrative values)

```
HTTP/1.1 422 Unprocessable Entity
Content-Type: application/json

{
  "error_code": "PAY-401",
  "message": "Request validation failed",
  "trace_id": "trc-000002"
}
```

Example: processor timeout (illustrative values)

```
HTTP/1.1 504 Gateway Timeout
Content-Type: application/json

{
  "error_code": "PAY-504",
  "message": "Payment processor did not respond in time",
  "trace_id": "trc-000003"
}
```

5.2 Reading Error Codes

Always use Table 5-1 to read a PAY code together with its HTTP status. The number in a PAY code is the Payments Team's own numbering; the HTTP status of each code is the one listed in the table.

Error code	Same body sent again	Client action
`PAY-400`	Fails again	Fix the JSON body.
`PAY-401`	Fails again	Fix the fields so the body matches payment-request/2.0.
`PAY-402`	Declined again	Ask the customer for another payment method.
`PAY-409`	Fails again	Use the original body for this idempotency key.
`PAY-504`	May succeed	Retry with the same idempotency_key.

Table 5-2. What happens if the same body is sent again.

6. Change History

Version	Date	Change
1.0	2025-03-10	First version. Field card_token.
2.0	2026-08-14	Renamed card_token to payment_method_token. card_token kept as deprecated alias.

Table 6-1. Specification versions.

6.1 Field Names by Version

Purpose	Spec 1.0	Spec 2.0
Payment method token	`card_token`	`payment_method_token (card_token accepted as deprecated alias until 2027-03-31)`
Order	`order_id`	`order_id`
Amount in minor units	`amount_minor`	`amount_minor`
Currency	`currency`	`currency`
Idempotency	`idempotency_key`	`idempotency_key`

Only the name of the payment method token field changed between version 1.0 and version 2.0.

7. Client Checklist

Use this checklist when building or changing a client of POST /v2/payments.

#	Check	Section
1	Send Content-Type: application/json and a valid JSON body.	2
2	Send all five required fields: order_id, amount_minor, currency, payment_method_token, idempotency_key.	3
3	Send amount_minor as an integer in minor units (4999 = 49.99).	3.1
4	Never send a raw card number. payment_method_token is a token.	3.1
5	Use one idempotency_key per order and reuse it on every retry of that order.	3.1
6	Use payment_method_token. card_token is deprecated and accepted only until 2027-03-31.	4
7	Read every error with Table 5-1: check both error_code and HTTP status.	5
8	Keep the trace_id of failed calls; it links the response to log records.	5.1

Table 7-1. Client checklist.

8. Glossary

Term	Meaning
Alias	An old field name that payment-service treats as the new name.
Deprecated	Still accepted for now, but planned for removal. Clients should stop using it.
Idempotency key	A value unique per order. Sending it again on a retry stops the payment from being recorded twice.
Minor units	The smallest unit of a currency, such as cents. 4999 = 49.99.
Schema	The list of fields, types and required flags a request must follow (here payment-request/2.0).
Trace ID	ID returned with every error. It links the response to log records.